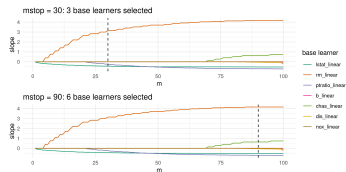
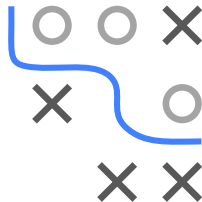


Introduction to Machine Learning

Boosting

Gradient Boosting: CWB Basics 1



Learning goals

- Concept of CWB
- Which base learners do we use
- Built-in feature selection

COMPONENTWISE GRADIENT BOOSTING

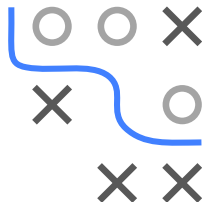
GB (with trees), has strong predictive performance but is difficult to interpret unless the base learners are stumps.

The aim of CWB is to find a model that exhibits:

- strong predictive performance,
- interpretable components,
- automatic selection of components,
- is sparser than a model fitted with maximum-likelihood estimation.

This is achieved by using “nice” base learners which yield familiar statistical models in the end.

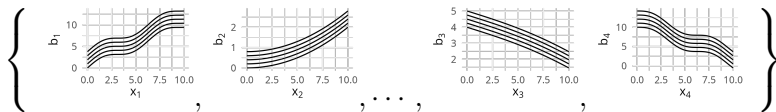
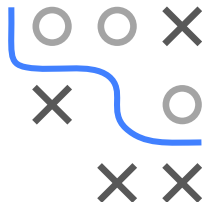
Because of this, CWB is also often referred to as **model-based boosting**.



BASE LEARNERS

In GB only one kind of base learner $\mathcal{B}$ is used, e.g., regression trees.

For CWB we generalize this to multiple base learner sets $\{\mathcal{B}_1, \dots, \mathcal{B}_J\}$ with associated parameter spaces $\{\Theta_1, \dots, \Theta_J\}$, where $j \in \{1, 2, \dots, J\}$ indexes the type of base learner.



In each iteration, base learners are fitted to the **pseudo residuals** $\tilde{r}^{[m]}$.

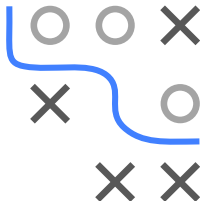
COMPONENTWISE BOOSTING ALGORITHM

Different from GB, multiple base learners $b_j \in \mathcal{B}_j, j = 1, \dots, J$, are fitted and only best-fitting one is selected and updated.

Algorithm Componentwise Gradient Boosting.

- 1: Initialize $f^{[0]}(\mathbf{x}) = \arg \min_{\theta_0 \in \mathbb{R}} \sum_{i=1}^n L(y^{(i)}, \theta_0)$
 - 2: **for** $m = 1 \rightarrow M$ **do**
 - 3: For all i : $\tilde{r}^{[m](i)} = - \left[\frac{\partial L(y, f)}{\partial f} \right]_{f=f^{[m-1]}(\mathbf{x}^{(i)}, y=y^{(i)}}$
 - 4: **for** $j = 1 \rightarrow J$ **do**
 - 5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:
 - 6: $\hat{\theta}_j^{[m]} = \arg \min_{\theta \in \Theta_j} \sum_{i=1}^n (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$
 - 7: **end for**
 - 8: $j^{[m]} = \arg \min_j \sum_{i=1}^n (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$
 - 9: Update $f^{[m]}(\mathbf{x}) = f^{[m-1]}(\mathbf{x}) + \alpha \hat{b}_{j^{[m]}}(\mathbf{x}, \hat{\theta}_{j^{[m]}}^{[m]})$
 - 10: **end for**
 - 11: Output $\hat{f}(\mathbf{x}) = f^{[M]}(\mathbf{x})$
-

(Same as for GB, New inner loop for CWB)



COMPONENTWISE BOOSTING ALGORITHM

Algorithm Componentwise Gradient Boosting (inner loop).

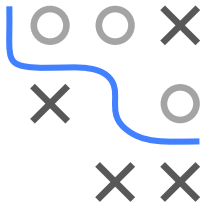
4: **for** $j = 1 \rightarrow J$ **do**

5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

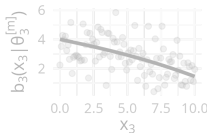
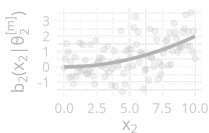
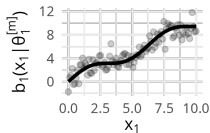
6: $\hat{\theta}_j^{[m]} = \arg \min_{\theta \in \Theta_j} \sum_{i=1}^n (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$

7: **end for**

8: $j^{[m]} = \arg \min_j \sum_{i=1}^n (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$



Iteration $m, j = 1, \sum_{i=1}^n (\tilde{r}^{[m](i)} - \hat{b}_1(x_1^{(i)}, \hat{\theta}_1^{[m]}))^2 = 24.4$:



COMPONENTWISE BOOSTING ALGORITHM

Algorithm Componentwise Gradient Boosting (inner loop).

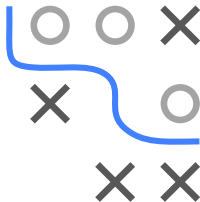
4: **for** $j = 1 \rightarrow J$ **do**

5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

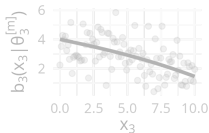
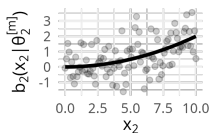
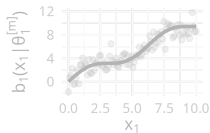
6: $\hat{\theta}_j^{[m]} = \arg \min_{\theta \in \Theta_j} \sum_{i=1}^n (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$

7: **end for**

8: $j^{[m]} = \arg \min_j \sum_{i=1}^n (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$



Iteration $m, j = 2, \sum_{i=1}^n (\tilde{r}^{[m](i)} - \hat{b}_2(x_2^{(i)}, \hat{\theta}_2^{[m]}))^2 = 43.2$:



COMPONENTWISE BOOSTING ALGORITHM

Algorithm Componentwise Gradient Boosting (inner loop).

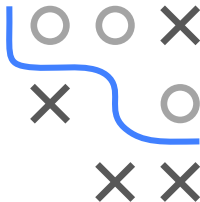
4: **for** $j = 1 \rightarrow J$ **do**

5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

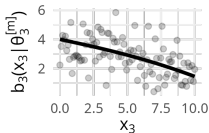
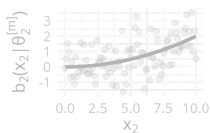
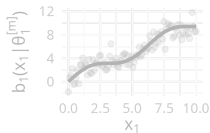
6: $\hat{\theta}_j^{[m]} = \arg \min_{\theta \in \Theta_j} \sum_{i=1}^n (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$

7: **end for**

8: $j^{[m]} = \arg \min_j \sum_{i=1}^n (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$



Iteration $m, j = 3, \sum_{i=1}^n (\tilde{r}^{[m](i)} - \hat{b}_3(x_3^{(i)}, \hat{\theta}_3^{[m]}))^2 = 35.2$:



COMPONENTWISE BOOSTING ALGORITHM

Algorithm Componentwise Gradient Boosting (inner loop).

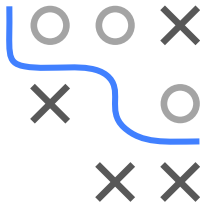
4: **for** $j = 1 \rightarrow J$ **do**

5: Fit regression base learner $b_j \in \mathcal{B}_j$ to the vector of pseudo-residuals $\tilde{r}^{[m]}$:

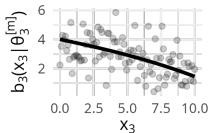
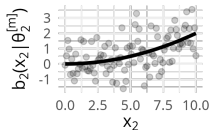
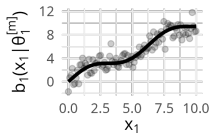
6: $\hat{\theta}_j^{[m]} = \arg \min_{\theta \in \Theta_j} \sum_{i=1}^n (\tilde{r}^{[m](i)} - b_j(\mathbf{x}^{(i)}, \theta))^2$

7: **end for**

8: $j^{[m]} = \arg \min_j \sum_{i=1}^n (\tilde{r}^{[m](i)} - \hat{b}_j(\mathbf{x}^{(i)}, \hat{\theta}_j^{[m]}))^2$



Iteration m : $\Rightarrow j^{[m]} = 1$



FEATURE SELECTION IN CWB

In CWB, we often define BLs on a single feature

$$b_j(x_j, \theta) \quad \text{for } j = 1, 2, \dots, p.$$

Allows natural form of feature selection:

- When we select the best BL in one iter of CWB, we thereby also only select one (associated) feature
- Note that a feature (or rather a BL associated with it) can be selected in multiple iters, so $\leq M$ features are selected

